



Aravindakshan R R 250701066-2025-CSE ▾

A2

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6      int arr[n];
7      for(int i = 0 ; i < n ; i++) {
8          scanf("%d", &arr[i]);
9      }
10     int k;
11     int i = 0, j = 1;
12     int found = 0;
13     scanf("%d", &k);
14     while( i < n && j < n ) {
15         int diff = arr[j] - arr[i];
16         if(diff == k && i != j) {
17             found = 1;
18             break;
19         } else if(diff < k) {
20             j++;
21         } else {
22             i++;
23         }
24     }
25     printf("%d", found);
26 }
```

Check

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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